Effect of Phase-Shifter Quantization (1-4 bits)

Same trained Transformer (multi-K_factor, K=1), evaluated at different hardware phase resolutions

1. Why This Test

Real SIM hardware can only set phase shifts to a limited number of discrete values, determined by the number of quantization bits (NbitQ): 2 bits = 4 possible phase values, 3 bits = 8, 4 bits = 16, etc. More bits means finer control but more expensive/complex hardware. This test measures how much achievable rate is lost (or gained) at each resolution, using the SAME trained Transformer model for all cases -- no retraining needed, since the model was trained on the continuous phase representation and only the final quantization step changes.

2. Results Summary

Bits (NbitQ)	Phase levels	Mean rate (bit/s/Hz)	Gain vs previous
1	2	8.38	--
2	4	16.41	+8.03 (+96%)
3	8	18.69	+2.28 (+14%)
4	16	19.16	+0.47 (+2.5%)

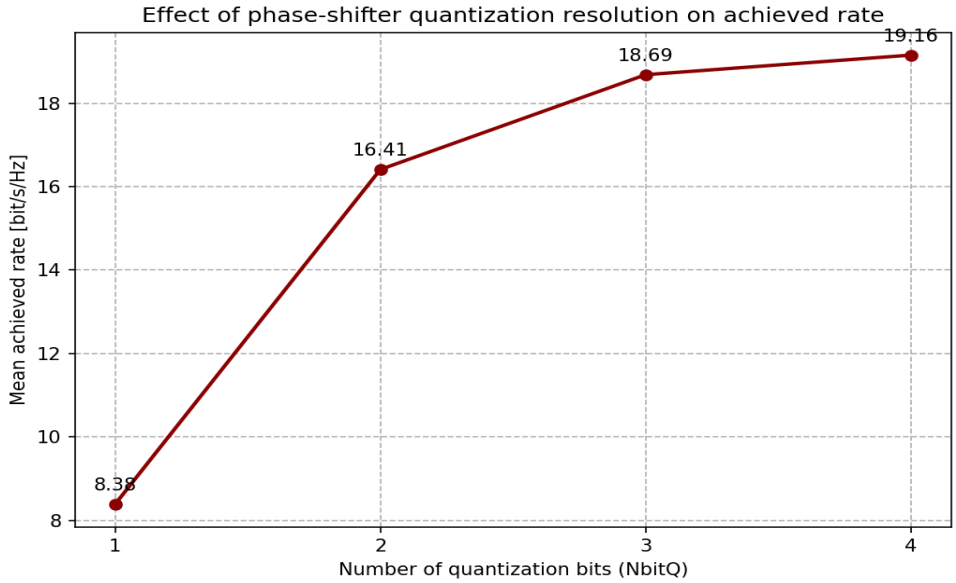


Figure 1. Mean achieved rate as a function of quantization resolution. Classic diminishing-returns curve: the jump from 1 to 2 bits nearly doubles the rate, while each additional bit beyond that brings progressively smaller gains, approaching the continuous (unquantized) ceiling of ~19.27 bit/s/Hz.

3. Interpretation

- **1 bit (2 levels)** is clearly insufficient -- only half the rate of 2 bits. Confirms that minimal hardware (simple on/off phase control) is a poor choice for this system.
- **2 bits (4 levels)** -- the configuration used throughout this project -- already captures most of the achievable gain (large jump from 1 bit).
- **3-4 bits** bring diminishing but still meaningful improvements, useful if the hardware cost allows it.
- Beyond 4 bits, further gains would be small -- the rate is approaching the continuous-phase ceiling, so there's a natural point of diminishing returns for hardware complexity vs. performance.

4. Files

Each bit depth has its own subfolder (bit1/, bit2/, bit3/, bit4/) containing the full results (.mat file with H, K_factor, predicted theta, and rate) and plots (rate vs K_factor, rate distribution), generated from the single shared trained model (transformer_multiKfactor_weights.pt).